

LeShun Palmprint Recognition - LS2 - Instructions for Use

1. Update log

Serial number	Update time	Document Version	AAR version	Update content	author
1	September 9, 2024	1.00	BaseLise:1.00 ShunPalm:1.00 ShunPalm-LS2:1.00	1. Create user documentation.	LiJUE
2	November 18, 2024	2.05	BaseLise:1.00 ShunPalm:2.00 ShunPalm-LS2:2.05	1. Update palm vein recognition algorithm to 1.3.10. 2. Increase memory loading feature values. 3. Add multi-threaded comparison of palm vein features.	LiJUE
3	December 03, 2024	2.07	BaseLise:1.00 ShunPalm:2.00 ShunPalm-LS2:2.07	1. Fix the problem where the model file address error caused the algorithm to fail to activate. 2. Fixed the issue of collection failure (147460) when continuously collecting palm prints.	LiJUE
4	December 23, 2024			English version	LiHanZhou

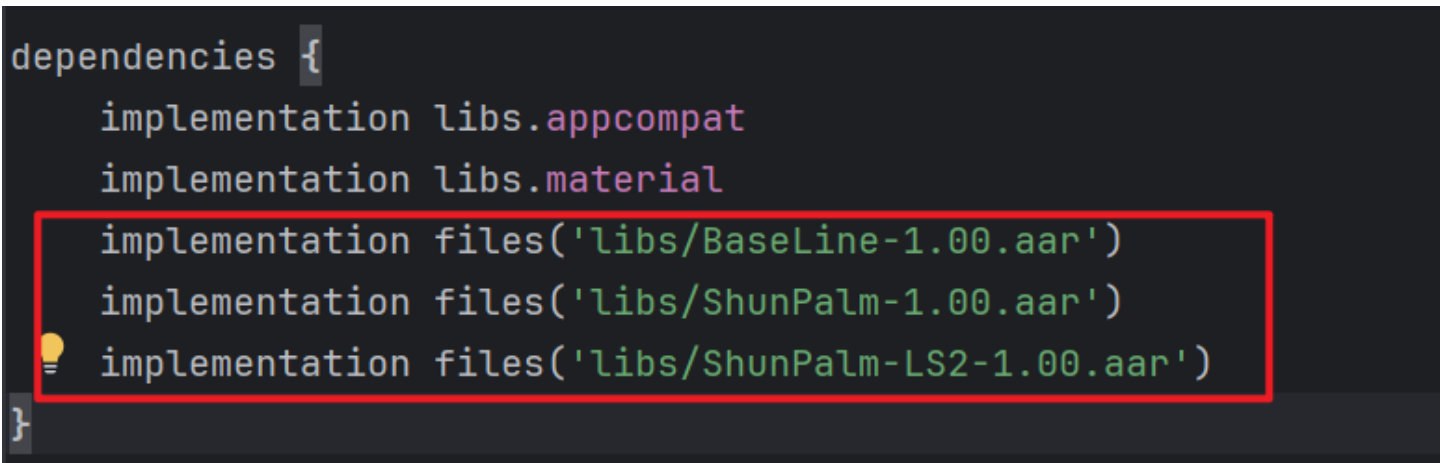
2. Access instructions

2.1 Import AAR package

1. Put the three files BaseLine-1.00.aar, ShunPalm-2.00.aar, and ShunPalm-LS2-2.07.aar into the libs folder.

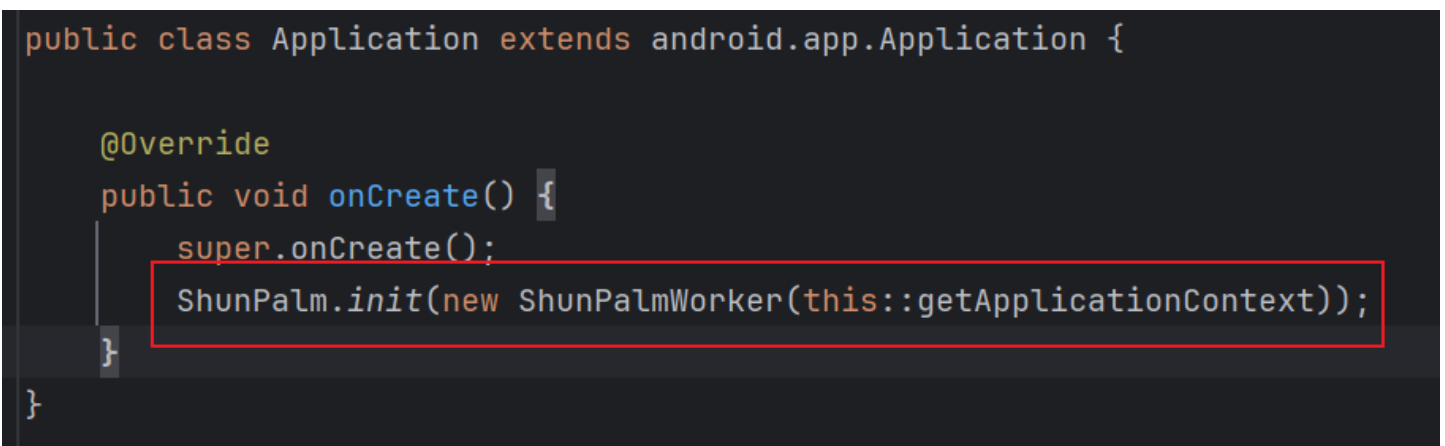


2. Add a reference to the build.gradle file.

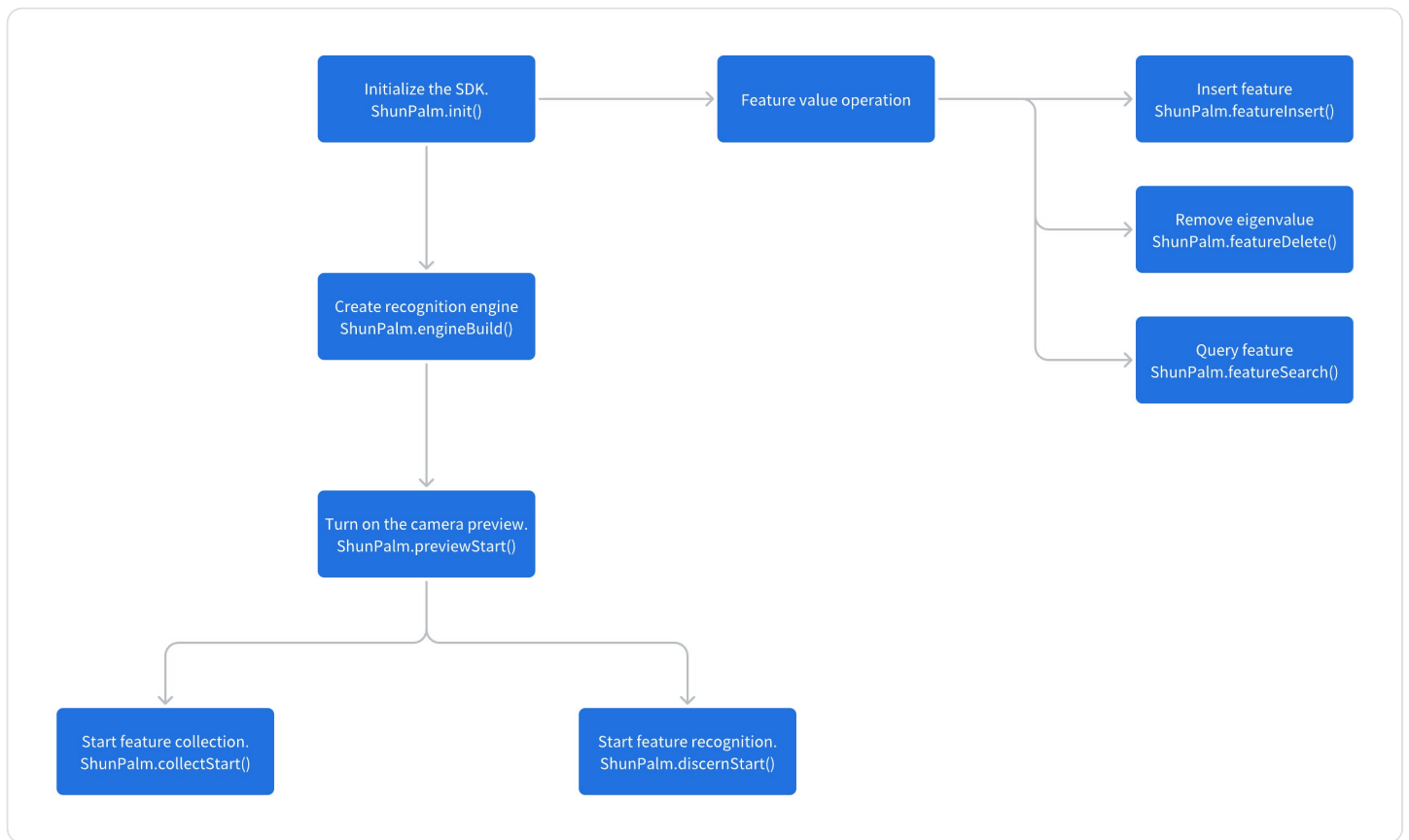


2.2 Initialize SDK

Call the ShunPalm.init () method in the onCreate method of the Application, passing in the com.leshun.support.shunplam.ls2.ShunPalmWorker class.



2.3 Method invocation flow



3. Method description

3.1 Initialize SDK

Method description:

This method is used to initialize the Palmprint Recognition SDK and must be called before calling all methods. The parameter passed to this method is `com.leshun.support.shunpalm.ls2.ShunPalmWorker`.

Method example:

```
1 // It is used to initialize the palmprint recognition SDK, and the calling method is as follows:
2 ShunPalm.init(new ShunPalmWorker(this::getApplicationContext));
```

3.2 Engine operation

3.2.1 Create recognition engine

Method description:

This method is used to create a palm print recognition engine.

Method example:

```

1  ShunPalm.engineBuild(new BaseLine.Params(), result -> {
2      switch (result.code()){
3          case WAITING:    // Engine is being created
4              break;
5          case SUCCEED:    // Engine created successfully
6              break;
7          case FAILURE:    // Engine creation failed
8              break;
9      }
10 });

```

3.2.2 Destroy identification engine

Method description:

This method is used to destroy the palm print recognition engine.

Method example:

```

1  ShunPalm.engineClear(result -> {
2      switch (result.code()){
3          case WAITING:    // Engine destroying
4              break;
5          case SUCCEED:    // Engine destroyed successfully
6              break;
7          case FAILURE:    // Engine destruction failed
8              break;
9      }
10 });

```

3.3 Recognition camera

3.3.1 Turn on the recognition camera

Method description:

This method is used to enable the recognition camera.

Method example:

```

1  ShunPalm.previewStart(new BaseLine.Params(), result -> {
2      switch (result.code()){
3          case WAITING:    // Camera on
4              break;

```

```

5         case SUCCEED:    // Camera turned on successfully
6             break;
7         case FAILURE:    // Camera failed to turn on
8             break;
9     }
10 });

```

3.3.2 Turn off the recognition camera

Method description:

This method is used to turn off the recognition camera.

Method example:

```

1  ShunPalm.previewCease(result -> {
2      switch (result.code()){
3          case WAITING:    // Camera is off
4              break;
5          case SUCCEED:    // Camera closed successfully
6              break;
7          case FAILURE:    // Camera shutdown failed
8              break;
9      }
10 });

```

3.4 Collect palm prints

3.4.1 Start collecting palm prints

Method description:

This method is used to start collecting palm prints

Method example:

```

1  ShunPalm.collectStart(new BaseLine.Params(), result -> {
2      switch (result.code()){
3          case WAITING:    // Palmprint collection is in progress.
4              break;
5          case SUCCEED:    // Palmprint collection was successful.
6              byte[] feature_rgb = result.get("feature_rgb", new byte[]{});
7              // Palmprint Features (RGB)
8              byte[] feature_nir = result.get("feature_rgb", new byte[]{});
9              // Palmprint Features (NIR)

```

```

8         int    palm_type    = result.get("palm_type", 0);        // Palm
    Type (0: Left Hand 1: Right Hand)
9         float  score        = result.get("score", 0.0f);        //
    Palmpoint quality
10        break;
11        case FAILURE:    // Palmpoint collection failed.
12        break;
13    }
14 });

```

3.4.2 Stop collecting palm prints

Method description:

This method is used to stop palm print collection.

Method example:

```

1  ShunPalm.collectCease(result -> {
2      switch (result.code()){
3          case WAITING:    // Stop collecting
4              break;
5          case SUCCEED:    // Stop collection successfully
6              break;
7          case FAILURE:    // Stop collection failed
8              break;
9      }
10 });

```

3.5 Recognize palm prints

3.5.1 Start recognizing palm prints

Method description:

This method is used to start recognizing palm prints

Method example:

```

1  BaseLine.Params params = new BaseLine.Params();
2  params.put("inquire_thread_size", 4);    // Palmpoint comparison thread
    count, type: int, default: 4 (1-8)
3  params.put("inquire_search_size", 2000); // Minimum contrast number of
    threads, type: int, default: 2000 (> 2000)

```

```

4  params.put("similar_rgb", 0.75);           // Palmprint contrast threshold,
    type: float, default: 0.75
5  params.put("similar_nir", 0.75);           // Palmprint contrast threshold,
    type: float, default: 0.75
6  ShunPalm.discernStart(params, result -> {
7      switch (result.code()){
8          case WAITING:    // Palmprint recognition is in progress.
9              break;
10         case SUCCEED:    // Palmprint recognition successful
11             ShunPalm.InquireData inquire_data = result.get("inquire_data",
ShunPalm.InquireData.class);
12             break;
13         case FAILURE:    // Palmprint recognition failed.
14             break;
15     }
16 });

```

3.5.2 Stop recognizing palm prints

Method description:

This method is used to stop recognition collection.

Method example:

```

1  ShunPalm.discernCease(result -> {
2      switch (result.code()){
3          case WAITING:    // Stop recognition
4              break;
5          case SUCCEED:    // Stop recognition success
6              break;
7          case FAILURE:    // Stop recognition failed
8              break;
9      }
10 });

```

3.6 Feature management

3.6.1 Insert palmprint feature

Method description:

This method is used to insert palmprint features into the palmprint library.

Method example:

```

1  String user_id = "user_id";    // user ID
2  String palm_id = "palm_id";    // Palmprint ID, non-repeatable
3  String voucher = "voucher";    // User verification code, when multiple
    users are identified, the user can be validated through this parameter
4  byte[] feature_rgb = new byte[]{};    // Palmprint Features (RGB)
5  byte[] feature_nir = new byte[]{};    // Palmprint Features (NIR)
6  int palm_type = 0;             // Palmprint type (0: left hand; 1: right
    hand)
7  float score = 0.0f;           // Palmprint quality
8  long code = ShunPalm.featureInsert(user_id, palm_id, voucher, feature_rgb,
    feature_nir, palm_type, result);
9  if (code >= 0) {
10     // User palmprint feature inserted successfully
11 } else {
12     // User facial feature insertion failed
13 }

```

3.6.2 Remove palmprint feature

Method description:

This method is used to remove palmprint features from the palmprint library.

Method example:

```

1  int size = ShunPalm.featureDelete(); // Remove all palmprint features
2
3  String user_id = "user_id";    // user ID
4  int size = ShunPalm.featureDelete(user_id);
5
6  String user_id = "user_id";    // user ID
7  String palm_id = "palm_id";    // Palmprint ID
8  int size = ShunPalm.featureDelete(user_id, palm_id);

```

3.6.3 Query palm print feature

Method description:

This method is used to query palmprint features from the palmprint library.

Method example:

```

1  List<ShunPalm.FeatureData> feature_list = ShunPalm.featureSearch(user_id);
2  List<ShunPalm.FeatureData> feature_list =
    ShunPalm.featureSearch(user_id, palm_id);

```

